

Air University Aerospace And Aviation Campus Kamra

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Lab Task 8

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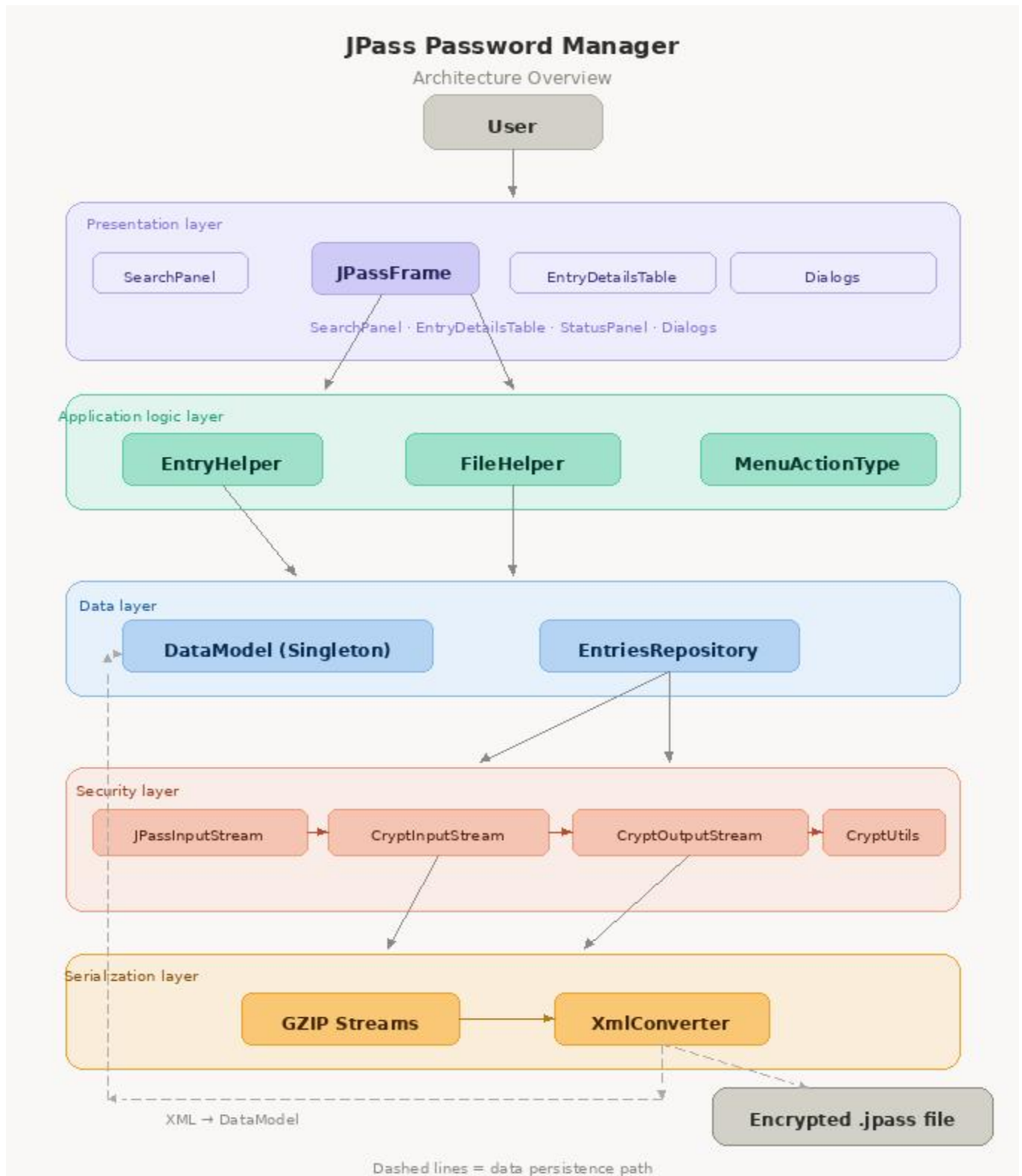
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Architecture Diagram



1. Secure Software Design Issues in JPass

JPass is a password manager, which makes it critical to ensure secure software design. The main design issues identified are:

1. Single Point of Failure

All passwords are stored in a single vault file (.jpass). If this file is compromised, all stored credentials are exposed.

2. Weak Encryption or Implementation Flaws

Improper encryption or weak key management could allow attackers to decrypt sensitive data. Strong AES encryption is essential for secure storage.

3. Master Password Dependency

The master password unlocks the entire vault. If the master password is weak or compromised, the attacker gains full access.

4. Insecure Storage or Serialization

JPass uses XML and GZIP for storage. Improper handling may lead to data leakage or integrity issues.

5. Device-Level Threats

Malware, keyloggers, or unauthorized software running on the user's device can capture passwords in transit or in memory.

6. Lack of Authentication Controls

Without multi-factor authentication, unauthorized access is easier if the master password is compromised.

2. Risk Assessment

Risk	Impact	Likelihood	Explanation
Master password compromise	High	High	All passwords are exposed if compromised
Encrypted vault theft	High	Medium	Offline brute-force attacks on the encrypted file
Malware or keylogger	High	High	Captures passwords as they are entered or used
Weak encryption design	Critical	Medium	Entire system can be broken if encryption fails
Data tampering	Medium	Medium	Data integrity can be lost during storage or transfer

3. Mitigation Strategies

1. Strong Encryption

- Use AES-256 for encryption.
- Apply proper key derivation functions such as PBKDF2 or bcrypt.
- Ensure end-to-end encryption for all sensitive data.

2. Master Password Policy

- Enforce minimum length and complexity requirements.
- Encourage passphrase-based passwords (e.g., Diceware).

3. Multi-Factor Authentication (MFA)

- Add OTP or biometric authentication to reduce risk from password compromise.

4. Secure Memory Handling

- Avoid storing passwords in plaintext in memory.
- Clear sensitive data from RAM after use.

5. File Protection and Integrity

- Implement HMAC for integrity verification.
- Detect tampering before loading stored data.

6. Protection Against Malware

- Auto-lock inactive sessions.
- Clear clipboard after use.
- Employ measures to prevent keylogging.

7. Secure Serialization

- Encrypt XML before GZIP compression.
- Avoid storing sensitive metadata unencrypted.

4. Architecture Improvements

To integrate these security measures, the architecture should be updated as follows:

1. Security Layer Enhancements

- Encryption Engine (AES-256)
- Key Derivation Function (PBKDF2)
- Integrity Checker (HMAC)

2. Authentication Layer

- Authentication Manager between User and Presentation Layer
- MFA handler for secure login

3. Data Layer Improvements

- Secure DataModel storing encrypted data in memory

4. Application Layer Enhancements

- Input validation module
- Secure session management (auto-lock, timeout)

5. Serialization Layer Enhancements

- XML encrypted before compression
- Integrity verification before file load

5. Conclusion

JPass faces critical risks from master password compromise, device-level attacks, and encryption weaknesses.

Mitigating these risks requires strong AES encryption, secure storage, and multi-factor authentication.

Incorporating an authentication layer and integrity checks into the architecture significantly enhances the overall security of the system.